

Effect of lignin removal on the properties of coconut coir fiber/wheat gluten biocomposite

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ABSTRACT

The effect of fiber lignin content on biocomposite properties was investigated. Coconut fiber was treated with 0.7% sodium chlorite to selectively decrease amounts of lignin. The fiber lignin content was then reduced from 42 to 21 wt.%. The composition and mechanical properties of the individual modified fibers were characterized. Gluten-based materials reinforced with modified fibers were prepared by compression molding. Then, the mechanical properties, water sensibility, matrix glass transition and infrared spectra of biocomposites prepared with fibers containing various amounts of lignin were evaluated. This study showed that the addition of coconut coir fiber significantly improved properties of wheat gluten biomaterials. In addition, the variation of lignin content in the fibers, in the investigated range, had no significant effect neither on matrix deplasticization nor fiber/matrix adhesion, suggesting that a partial lignin removal is not an efficient way to improve the properties of natural fiber/plasticized protein biocomposites.

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1. Introduction

The increase in fossil energy costs and the environmental concerns result in new opportunities for the industrial production of biodegradable materials based on natural renewable resources. A growing demand for various applications are thus expected such as short-lived applications for agriculture (e.g., plant pot, mulching films to cover soil), food and non-food packaging [1,2]. Gluten-based material displays interesting functional properties, in terms of viscoelasticity and water resistance. Mechanical properties of gluten-based materials can be modulated according to the process conditions for example temperature [3,4] or mechanical energy input [5], or to the blend composition by the modification of the plasticizer content [6] or by the addition of natural fibers [7–9].

Natural fibers, which are essentially composed of cellulose, hemicellulose and lignin, are widely used as a reinforcement to produce biocomposite [7–9]. The fiber composition depends on the plant from which it is extracted, as well as on the agricultural conditions. It is mainly composed of three compounds which are cellulose, hemicellulose and lignin. Cellulose and hemicellulose are polysaccharides, while lignin is a three-dimensional amorphous polyphenolic macromolecule consisting of three types of phenylpropane units (as shown in Fig. 1) [10], which are forming

a complex, highly branched and amorphous structure. Moreover, the local repartition of the compounds is not homogeneous. In general, lignin is mainly located at the surface of the fiber, while the backbone is mainly composed of cellulose.

Adhesion between matrix and fiber is an important parameter affecting the mechanical properties of composite, as a good adhesion ensures a good stress transfer from the matrix to the fiber [11]. This adhesion can result from a physical origin or from a chemical cross-linking. In natural fibers/wheat gluten composites, both types of adhesion are supposed to be effective. Our previous study [8] showed that different reinforcement effect can be correlated with different Pressure Sensitive Adhesive (PSA) properties of the gluten matrix. Additionally, chemical bonding can strongly affect the quality of the interface. Lignin, a polyphenolic compound located on fiber surface, may play a key role on the fiber/matrix chemical adhesion. Indeed, polyphenol/protein interactions have been largely described in literatures [12–14] and various types of interactions are identified. In a recent study, we have demonstrated that Kraft lignin can strongly interact with wheat gluten [15], and evidenced the role of the phenolic group in this interaction [16]. Therefore, variations in the fiber lignin content should monitor the density of fiber/matrix interactions, and resultantly the biocomposite properties.

A specific phenomenon that can be observed in biocomposites is called the matrix deplasticization [8]. Indeed, the agropolymer used as a matrix (here, a protein), is thermosensitive, and begins

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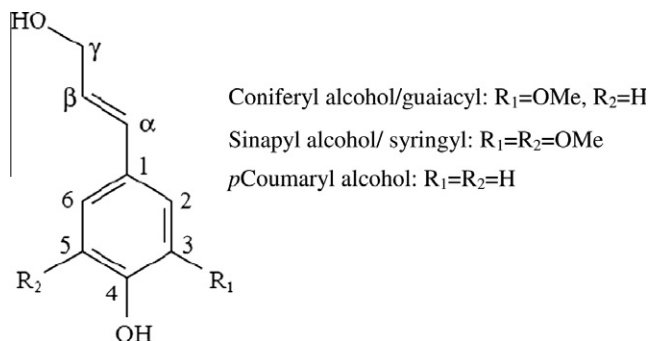


Fig. 1. The three building blocks of lignin [10].

to degrade at a temperature lower or close to its glass transition. Therefore, agropolymers need to be plasticized by a small polar molecule (as glycerol) to decrease their glass transition temperature, and therefore allow their industrial processing with a limited thermal degradation. This plasticizer significantly affects the matrix properties, the common observation is that it decreases both Young's modulus and tensile strength, while increasing the elongation at break of materials [8]. Then, when natural fibers are added, there is a competition between the matrix and the fibers for the plasticizer absorption, which can result in a deplasticization of the matrix, and thus in a different reinforcing effect. Unlike cellulose which is associated in microfibrils, lignin is an amorphous polymer, and thus may play an important role on this mechanism. Therefore, lignin might play a key role by increasing the mechanical properties of those biocomposites due to its location on the surface fibers, its amorphous structure, and its reactivity with wheat gluten.

Coconut coir fiber, which is in average composed of 46% of lignin (weight basis) is one of the natural fibers containing the higher lignin content [17]. Lignin can be extracted selectively and progressively by treating the fibers with an aqueous alkaline solution or with an organic solvent [2]. It is thus a medium to conduct a systematic study on the effect of lignin on biocomposites properties. About 55 billion of coconuts are harvested annually in the world, but only 15% of the husk fibers are actually recovered for use [18]. Most husks are abandoned in the nature, which constitute a waste of natural resources and a cause of environmental pollution [19]. Therefore, biocomposites from wheat gluten reinforced with coconut coir fiber would certainly offer interesting routes for the production of environmentally-friendly materials. Use of coconut coir as a reinforcement has been already studied, but only on cement board [20], polypropylene [21], and starch/ethylene vinyl alcohol copolymers [22].

Pretreatments of coir fiber by washing and boiling in order to remove the impurities on the coir surface have been already studied [20]. In terms of surface topology, pretreatments can create voids and produce fiber fibrillation, leading to a better fiber/matrix adhesion and therefore better mechanical properties of coir/cement composite. For biocomposite, the modification of fiber chemical composition and their effect on the properties of materials was studied [23,2]. The effect of the lignin content has been studied on lignocellulosic fibers incorporated into a biodegradable aromatic polyester, polybutylene adipate-co-terephthalate [2]. In that case, lignin removal by chemical treatment increased the biocomposite moduli, suggesting that the lignin/cellulose ratio is an important parameter [2]. However, the effect of chemical composition of natural fiber on properties of biocomposite especially on matrix deplasticization and fiber/matrix interaction has not been clearly reported.

The objective of this work was to study the reinforcing effect of coconut fiber in protein-based biocomposites, by modifying the fiber lignin content. Firstly, coconut coir fiber was pretreated in order to decrease progressively its lignin content. Properties of original and modified fibers were characterized. Then, the glass transition temperature (T_g) of biocomposite was determined by dynamic mechanical thermal analysis (DMTA), in order to investigate the matrix deplasticization. Chemical bonding between fibers and matrix were investigated by Fourier Transform Infrared Spectroscopy (FTIR). Mechanical properties and water absorption of the samples were finally characterized to study the functional properties of materials.

2. Experimental procedure

2.1. Materials

Commercial vital wheat gluten was obtained from Winner Group Enterprise Ltd. (NSW, Australia). Its protein content was 76.8% (dry matter), moisture content was 6% (wet basis) according to the manufacturer.

Coconut coir fibers were purchased from Banglamung factory (Chonburi, Thailand). They are obtained by separating fiber and pitch, and drying in an ambient air. Density of raw coconut fiber is $0.86 \pm 0.06 \text{ g/cm}^3$ measured by oil pycnometer.

Anhydrous glycerol was purchased from Roongsub Chemical Ltd. (New South Wales, Australia) in analytical grade. Chemical reagents were obtained from Ajax Finechem Ltd., Merck and Carlo Erba Ltd. in analytical grade.

2.2. Fiber preparation and characterization

2.2.1. Lignin extraction

The selective extraction of lignin from coconut fibers was realized as follow [24]. Fibers were treated with 0.7% NaClO_2 at pH 4, 1:50 liquor ratio, at boil temperature for different periods of time at 0, 15 and 90 min. After treatment, fibers were washed, dried at 105°C for 5 h and cut into 5 mm length.

2.2.2. Chemical composition and crystallinity of fiber

Moisture content and ash content of coconut fiber were analyzed by AOAC (2006) 990.19, 945.46. Lignin content, holocellulose content and alpha-cellulose content of coconut fiber were analyzed by Klason method, browning method and TAPPI T201 om-93, respectively. Hemicellulose content was calculated by the difference between holocellulose and alpha-cellulose. Samples were analyzed in three replications.

Crystallinity of cellulose was analyzed using X-ray diffractometer (Rigaku, DMAX 2200). X-ray diffraction spectra were collected in a 2θ range between 10° and 50° . The crystallinity index, which is adapted only to crystalline cellulose I, was calculated by [25]:

$$\text{Crystallinity (\%)} = (I_{002} - I_{18.5^\circ}) / I_{002} \times 100 \quad (1)$$

where I_{002} is the diffracted intensity by (0 0 2) plane which is the intensity at strongest peak [26], $I_{18.5^\circ}$ is the diffracted intensity at 18.5° .

2.2.3. Mechanical properties of single fiber

The fiber tensile strength test was carried out by using a Texture Analyzer (Stable Micro System, TA-XT.plus, Surrey, UK). Coconut coir fibers were cut into 9 cm length. The diameters of coconut fibers are between 0.289–0.577 mm, 0.226–0.637 mm and 0.236–0.497 mm measured using a caliper for 0, 15 and 90 min of treatment, respectively. The initial grip separation was 50 mm and elongation speed was 1 mm/s. Stress values (MPa) were

calculated by dividing the measured force values (N) by the initial cross-sectional area of the specimen (mm^2). Strain values were expressed in percentage of the initial length of the elongating part of the specimen ($L_0 = 50$ mm). Young's modulus was determined as the slope of the linear regression of the stress–strain curve. Samples were analyzed in 20 replications.

2.2.4. Surface morphology of fiber

Surface morphology of the fibers was examined by scanning electron microscopy each sample was deposited on carbon tape mounted on stubs and then gold-coated. Samples were observed by scanning electron microscopy (Hitachi S-3400N & EDAX) using a voltage of 10 kV.

2.3. Biocomposite preparation

The composites were produced by mixing and compression molding. In order to decrease the gluten glass transition temperature and therefore process it at a processing temperature that prevented a strong thermal degradation, glycerol was added. Composition consisted of 65 gluten/35 glycerol wt./wt. as a matrix, and 10 wt.% fiber as a reinforcement.

2.3.1. Mixing process

Sixty gram of gluten, glycerol and fiber was mixed in a mixer (King Mixer, K-05 model, US). Three steps of mixing speed were successively used for each sample: low, medium and high, respectively. Mixing time of each speed is 10 min.

2.3.2. Compression molding

Fifty-five gram of the mixed blend was deposited in a squared mould ($15 \text{ cm} \times 15 \text{ cm} \times 2 \text{ mm}$) and thermomolded at 130°C , 150 bar in a heated press (LP-25M, Labtech Engineering Co., Ltd., Thailand) for 15 min.

2.4. Biocomposite characterization

2.4.1. Mechanical properties at high deformation

Tensile tests were performed on a Texture Analyzer (Stable Micro System, TA-Xt.plus, Surrey, UK). Samples were cut into dumb-bell shaped specimens of 11 cm overall length and 7 mm width by Hydraulic Press Machine (15 T., SMC TOYO METAL Co., Ltd., Thailand) and preconditioned at 25°C and 53% relative humidity over a saturated salt solution of $\text{Mg}(\text{NO}_3)_2$. Specimen thickness was measured with a caliper. The initial grip separation was 50 mm and elongation speed was 1 mm/s. Tensile strengths and elongations at break values, as well as Young moduli were calculated as described in Section 2.2.3. Each sample was analyzed at least in four replications.

2.4.2. Mechanical properties at low deformation

Rectangular samples ($10 \times 3 \times 1 \text{ mm}^3$) were analyzed with a dynamic mechanical thermal analyzer (NETZSCH DMA 242, Piscataway, USA) equipped with a cryogenic system fed with liquid nitrogen. A tensile test was performed with a temperature ramp from -100 to 150°C at a heating rate of 3°C min^{-1} . A variable sinusoidal mechanical stress was applied to the sample (frequency = 1 Hz) to produce a sinusoidal strain amplitude of 0.05%, which ensures measurements in the linear domain of viscoelasticity. During analysis, the storage modulus (E'), the loss modulus (E'') and $\tan \delta$ ($=E''/E'$) were recorded and plotted against temperature for further evaluation of thermal transition. T_g was identified as the temperature of the $\tan \delta$ maximum. Each sample was analyzed in three replications, and the average value is given.

2.4.3. Water absorption

Samples (20 mm in diameter) were dried in hot air oven at 55°C until their weight was constant (W_i). Then, they were immersed in 50 ml distilled water containing 0.05% NaN_3 to avoid the microbial growth at 25°C . The swollen samples were wiped and weighed (W_w) after 1 week. Then, they were dried in hot air oven at 55°C until their weight was constant (W_f). Each sample was analyzed in four replications.

$$\text{Water absorption (\%)} = 100 \times (W_w - W_f) / (W_i) \quad (2)$$

2.4.4. FTIR

The composites were sprinkled into a matrix of KBr, and ground in an agate mortar (KBr pellet technique). The samples were tested using a Fourier Transform Infrared Spectrometer (Perkin Elmer instruments, Singapore). Investigation had been performed in the transmission mode at the resolution of 4 cm^{-1} . Each sample recording consisted of 64 scans recorded from 400 to 4000 cm^{-1} .

2.5. Statistical analysis

The data in this experiment were analyzed and presented as mean values with standard deviations. Differences between mean values were established using one-way analysis of variance (ANOVA) and least significant difference (LSD) test at 95% confidence ($p < 0.05$). The software SPSS 10.0 program was used to perform the calculation.

3. Results and discussion

3.1. Influences of lignin extraction on fiber properties

3.1.1. Chemical composition

The chemical compositions of unextracted and extracted coconut fibers are given in Table 1. Unextracted coconut fiber has a lignin content about 42%. After lignin extraction for 15 and 90 min, fibers contain respectively 31% and 21% of lignin, corresponding to 25% and 50% of lignin removal. Therefore, sample codes of fiber in Table 1 which are L 42, L 31 and L 21 correspond to lignin content after 0, 15 and 90 min extraction, respectively. As result of the sodium chlorite (NaClO_2) treatment and resulting lignin extraction, the relative cellulose content of the fibers increase. However, only slight changes in cellulose crystallinity are observed.

3.1.2. Fiber surface

The NaClO_2 treatment used in this study to remove lignin from the fiber surface also removes impurities, wax and fatty substances presented on the fiber surface. It can be observed in Fig. 2 that the longer treatment results in the stronger effect. The globular protrusions composed of fatty deposits called tylose [27], have disappeared. Similarly, the rugosity associated with the presence of parenchyma cells [28] at the fiber surface is reduced. Thus, fiber surface appears to be smoother.

Table 1
Chemical composition of coconut fiber extracted with different time.

Lignin extraction time (min)	Sample code of coconut fiber	Crystallinity of cellulose (%)	Chemical composition (%)		
			Lignin	Cellulose	Hemicellulose
0	L 42 fiber	51.98	42.10	32.69	22.56
15	L 31 fiber	52.59	31.38	37.67	24.63
90	L 21 fiber	49.06	21.42	43.75	24.83

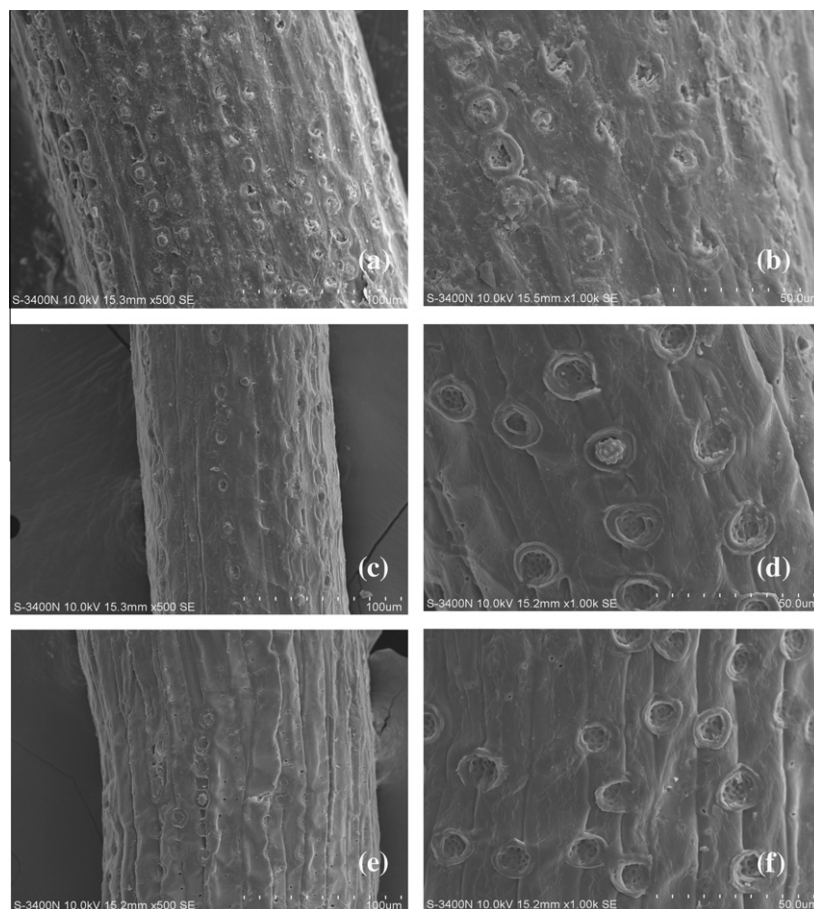


Fig. 2. Coconut fiber, 500 $\times$ (a), 1000 $\times$ (b), 15 min- NaClO_2 treated fiber, 500 $\times$ (c), 1000 $\times$ (d), and 90 min- NaClO_2 treated fiber 500 $\times$ (e), 1000 $\times$ (f).

Table 2
Mechanical properties of coconut fiber with different treatment time.

Coconut fiber	Young's modulus (GPa)	Tensile strength (MPa)	Elongation at break (%)
L 42 fiber	2.29 ± 0.47^a	123.2 ± 34.7^b	33.39 ± 7.01^c
L 31 fiber	2.59 ± 0.64^a	97.3 ± 37.4^a	21.61 ± 9.00^a
L 21 fiber	2.43 ± 0.62^a	112.5 ± 47.8^{ab}	27.59 ± 11.95^b

Values with different superscript letters in the same column are significant difference ($p < 0.05$). Significance of the differences was tested with ANOVA–LSD test as described in Section 2.5.

3.1.3. Mechanical properties of fiber

Mechanical properties of the fibers containing different lignin content are shown in Table 2. Mechanical properties of unextracted fiber (L 42 fiber) are closed to the values in literature which reported Young's modulus of 2.2–2.4 GPa, tensile strength of 128–155 MPa and elongation at break of 28–34% [29].

Table 2 shows that the elongation at break and tensile strength of the fibers decrease to a minimum and then increase again while increasing amounts of lignin removal. Indeed, the effect of lignin extraction on fibers properties can be diverse. On one hand, the partial breakage of the three dimensionally cross-linked network of cellulose and lignin after treatment, results in lower adhesion within the fiber [30], and thus in lower mechanical properties. But on the other hand, the removal of cementing materials affects the rearrangement of the cellulose molecules, leading to a better packing of cellulose [25]. The extraction can also modify the fibers dimensions and diameters, which is known to modify their tensile strength properties, which increase when the diameter decreases

[31], as measured in this study for the stronger treatment (Section 2.2.3). Table 2 also shows that Young's modulus and tensile strength of untreated and treated fibers are not significantly different. Therefore, using those different fibers in biocomposites allow us to vary their chemical compositions while preserving their elastic properties.

3.2. Effect of lignin content on composite properties

Gluten materials reinforced with 10 wt.% unextracted and extracted coconut fiber were prepared. Density of coconut fiber is about $0.86 \pm 0.06 \text{ g/cm}^3$, which is low compared to another fiber such as hemp fiber which has density of 1.44 g/cm^3 [32]. Previous study [8] showed that 20 wt.% or 18 vol.% of hemp fiber caused the fiber agglomeration in gluten-based biocomposite. Therefore, in this study 10 wt.% (or 14 vol.%) of coconut fiber was selected to avoid an agglomeration of fiber in the matrix.

3.2.1. Mechanical properties of composite

The effect of lignin content on mechanical properties of fiber/gluten composite is shown in Table 3. The addition of fibers results in a strong increase of Young's modulus, an increase of the tensile strength, and a decrease of the elongation at break, as usually observed in reinforcement. This is typical of a reinforcing effect, and those data are similar to values already published for gluten/fiber composite [9]. Young's modulus increase with the addition of 10% of fibers is about 3–4 times. For a given fiber content, composites properties depend on the fibers individual properties, and on the matrix/reinforcement adhesion. Table 3 shows that the elongation at break of the composite follows the same trend as that

Table 3

Mechanical properties of gluten-based material reinforced with 10% coconut fiber containing different lignin content.

Sample	Young's modulus (MPa)	Tensile strength (MPa)	Elongation at break (%)
Gluten-based material	5.52 ± 0.55	1.71 ± 0.12	162.7 ± 25.1
<i>Gluten/fiber composite</i>			
L 42 fiber	17.94 ± 2.26 ^a	1.86 ± 0.13 ^a	32.82 ± 6.07 ^a
L 31 fiber	22.74 ± 2.54 ^b	1.85 ± 0.11 ^a	23.00 ± 6.40 ^b
L 21 fiber	18.51 ± 1.40 ^{ab}	1.76 ± 0.05 ^a	29.53 ± 2.92 ^a

Values with different superscript letters in the same column are significant difference ($p < 0.05$). Significance of the differences was tested with ANOVA–LSD test as described in Section 2.5.

of the individual fibers, while tensile strength does not show any significant difference. Young's modulus of the composites slightly increases when the lignin content is reduced. However, changes of composite mechanical properties with lignin content remain very limited in comparison with the global reinforcing effect due to the fiber addition. In the investigated range, the partial lignin removal does not appear as strongly modify the biocomposite properties.

In this study, the lignin content is reduced from 42 to 21 wt.%. Beside this important difference in composition, the remaining lignin content might still be sufficient to cover the fiber surface. Therefore, it seems that in natural fiber/protein biocomposite, the fiber lignin content does not affect the fiber/matrix adhesion as long as it is still sufficient to cover the fiber surface.

3.2.2. Glass transition temperature

Storage modulus (E') and $\tan \delta$ evolutions with temperature of composites reinforced with 10% fiber containing different lignin content are shown in Fig. 3. Several $\tan \delta$ peaks are observed. A minor peak at a temperature range from -50 to -60 °C should be attributed to the second relaxation ($\tan \delta - \beta_1$) of free glycerol [33]. The weak peak ($\tan \delta - \alpha_2$) appearing at a temperature around 26 – 33 °C should be attributed to the T_g of the plasticized matrix phase [34]. A strong $\tan \delta$ peak ($\tan \delta - \alpha_1$) is corresponded to the main transition of material.

Table 4 shows the T_g and $\tan \delta$ peak of materials determined from the main transition. Fiber addition increases T_g of material compared to pure gluten-based material. This can be associated with a deplasticizing effect [9] and/or interaction between fiber and matrix [35] as already reported. Moreover, when fiber was

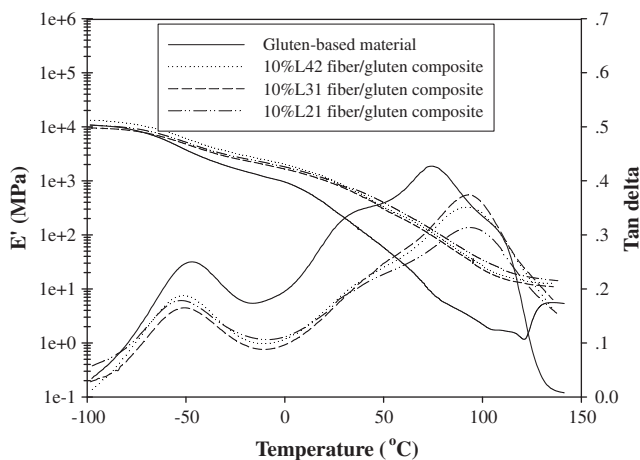


Fig. 3. E' and $\tan \delta$ of wheat gluten-based materials reinforced with coconut fiber containing different lignin content.

Table 4

T_g and $\tan \delta$ peak height of composites with different lignin content.

Sample	T_g (°C)	$\tan \delta$ peak height
Gluten-based material	72.20 ± 3.0	0.43 ± 0.0047
<i>Gluten/fiber composite</i>		
L 42 fiber	92.37 ± 0.8	0.35 ± 0.0014
L 31 fiber	92.51 ± 0.1	0.36 ± 0.0202
L 21 fiber	92.33 ± 1.6	0.31 ± 0.0004

added into material, peak at a temperature around 26 – 33 °C disappears. The maximum of $\tan \delta$ of pure gluten material is lower than that of fiber/gluten composite, suggesting motional restriction [34] due to fiber addition.

Concerning the effect of lignin content on T_g and $\tan \delta$, lignin removal has no significant effect neither on T_g or $\tan \delta$ peak. This suggests that the matrix deplasticization can originate in the plasticizer absorption either by the cellulose, the hemicellulose or the lignin, in similar proportions. As a result, the addition of a similar concentration of fibers, whatever their relative lignin content results in a similar deplasticizing effect.

3.2.3. Water absorption

Fig. 4 shows the water absorption of gluten-based material and gluten-based materials reinforced with 10% fiber containing different lignin content. The addition of 10% fibers strongly reduces the water absorption of the materials from 75% to 66.5%. These values are closed to the ones found in literature for gluten/fiber composite [9]. The overall water absorption of a sample is simply the sum of the water absorption of each of its components, balanced by their weight fraction. Therefore, in average, fibers absorb less water than the plasticized wheat gluten matrix. Fig. 3 shows that lignin removal slightly decreases the water absorption of composite. This result shows that a difference exists between the water absorption of lignin and of cellulose (and hemicellulose). As lignin is easily accessible and has an amorphous structure, it can absorb more water than cellulose, which is crystalline and less accessible [24,36].

3.2.4. FTIR spectra

To study the reinforcing effect of fibers with variables lignin content, the FTIR spectra of composites was characterized to observe the formation of new chemicals bonds between fibers and gluten. Fig. 5 shows the FTIR spectra from 1600 to 1000 cm^{-1} of the WG-based material with fibers containing different lignin content. Observed peaks are the functional groups of gluten or fiber. There are some slight changes in band positions and intensities.

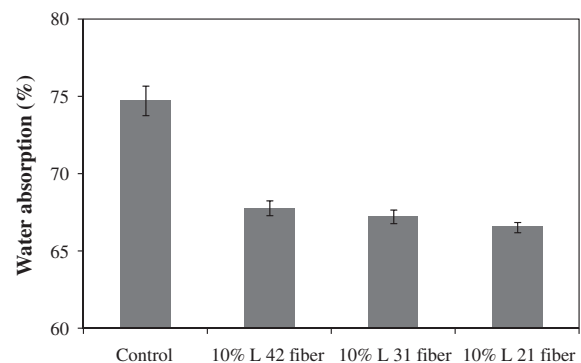


Fig. 4. Water absorption of gluten-based material and gluten-based materials reinforced with 10% fiber containing different lignin content.

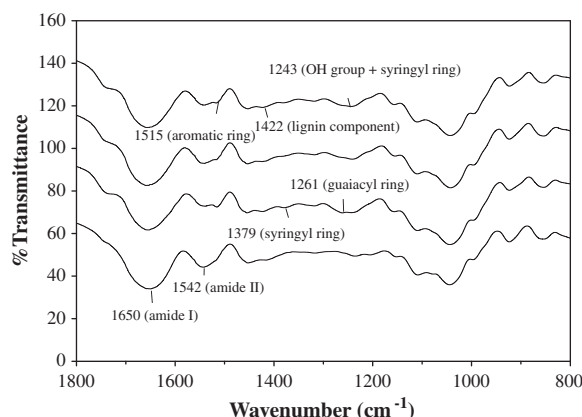


Fig. 5. FTIR spectra (bottom to top) of gluten-based material, and gluten-based composites with 10% of L42, L31 and L21 fiber.

All materials present peaks at 1640 cm^{-1} (amide I) and 1542 cm^{-1} (amide II) which are functional group of gluten. For fiber/gluten composite, some peaks characteristics of the fibers can be clearly observed, for example at 1515 cm^{-1} (aromatic ring), at 1422 cm^{-1} (lignin component), and at 1243 cm^{-1} (OH group and syringyl ring). However, no new chemical bond between fiber and matrix is observed. Moreover, it is difficult to deduct from those measurements the complete absence of those chemical bonds, as their absence may be due to a very complex infrared spectrum of both lignin and gluten, which have a large number of functions and linkages in their structure.

4. Conclusion

In this study, the results showed that the properties of coconut coir/wheat gluten biocomposites are significantly different from those of pure plasticized gluten materials. Up to 50% lignin content of the fibers was progressively removed. Then, the effect of this composition change was evaluated for lignin content ranging between 42 and 21 wt.% in the fibers. In this range, lignin removal does not modify the mechanical properties of coconut fiber itself. In terms of reinforcing effect, matrix deplasticization or overall biocomposite mechanical properties, the lignin removal has no significant effect, but slightly reduces the water absorption of samples.

The hypothesis is that the remaining lignin is still sufficient to cover the fiber surface, where it is essentially located. Therefore, this study suggests that in natural fiber/protein biocomposites, a high lignin content in the fibers is not a necessary condition to obtain a good fiber/matrix adhesion, at least if the lignin concentration is sufficient to cover fiber surface. However, a definitive demonstration of this hypothesis will imply the preparation fibers in which the surface lignin content can be modulated precisely from nothing to a complete coverage. Therefore, further studies will be conducted in order to develop lignin extracting procedures that will allow reaching higher extraction level without degrading the inner structure of natural fibers. Evidencing the condition in which lignin content affects or not the biocomposite properties leads to a selection of appropriate fiber and fiber treatment to biocomposite production.

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